CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

# Chapter 6: We Distribute, Yet Things Multiply

Every formula and result of the chapter on one sheet

| | | |
|---|---|---|
| 1 | Distributive property | $a(b+c) = ab + ac$ |
| 2 | Product of two brackets | $(a+m)(b+n) = ab + an + mb + mn$ |
| 3 | Square of a sum | $(a+b)^2 = a^2 + 2ab + b^2$ |
| 4 | Square of a difference | $(a-b)^2 = a^2 - 2ab + b^2$ |
| 5 | Difference of squares | $(a+b)(a-b) = a^2 - b^2$ |
| 6 | Sum of the two squares | $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ |
| 7 | Difference of the two squares | $(a+b)^2 - (a-b)^2 = 4ab$ |
| 8 | Śrīdharācārya's square | $a^2 = (a+b)(a-b) + b^2$ |
| 9 | Cube of a sum | $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ |
| 10 | Difference of cubes | $(a-b)(a^2 + ab + b^2) = a^3 - b^3$ |
| 11 | Both numbers increased by 1 | $(a+1)(b+1) = ab + a + b + 1$ |
| 12 | Multiplying by 11 | $N \times 11 = 10N + N$ |
| 13 | Multiplying by 99 | $N \times 99 = 100N - N$ |
| 14 | Consecutive numbers | $n^2 - (n-1)(n+1) = 1$ |